

The effect of bond impurity in triangular lattice heisenberg anti-ferromagnet

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Abstract

In this note we explore the fate of the 120 degree long range order in the presence of randomly distributed bond impurities.

1 Introduction

In the context of the classical spiral ground state of the triangular lattice heisenberg anti -ferromagnet, the presence of a single bond impurity locally releases the frustration and helps the associated spins to anti-align. In the reference frame of an individual spin this amounts to a perturbation in the transverse direction of the magnetic order. If we assume a rotated co-ordinate system for each spin such that in the absence of any fluctuation, $\langle \tilde{S}^z \rangle = S$, the effect of a single impurity could be understood as a perturbation to the hamiltonian,

$$\delta H_i = h_i \sum_{j \in \{i\}} \alpha_j \tilde{S}_j^x = h_i S_i^{\text{imp}} \quad (1)$$

where the notation is as follows; the bond impurity has been introduced between the coupling of the i 'th spin and one of its neighbour and j is either 0 or 1 depending on whether it designates the i 'th spin or its neighbour. The phase factor $\alpha_j = \pm(-1)^j$ ensures the correct symmetry of the perturbation. Because of the spiral pattern of the underlying spin pattern the neighbouring spins have to rotate in the opposite directions in-order to anti-align. The overall $\pm$ of the phase factor is fixed with the breaking of the chiral symmetry of the 120° order. However, this does not have any effect on the observables that we are interested in.

2 The response function

Within the linear response approximation the transverse spin fluctuation of an arbitrary spin j away from the impurity at i would be given by,

$$\begin{aligned}
\langle \tilde{S}_j^x \rangle &= h_i \langle \tilde{S}_j^x S_i^{\text{imp}} \rangle = h_i \langle \tilde{S}_j^x \sum_{j' \in \{i\}} \alpha_{j'} \tilde{S}_{j'}^x \rangle \\
&= h_i \sum_{\vec{k}, \vec{k}'} \langle \tilde{S}_{\vec{k}}^x \left(\sum_{j' \in i} \alpha_{j'} e^{i\vec{k}' \cdot \vec{r}_{j'}} \right) \tilde{S}_{\vec{k}'}^x \rangle e^{i\vec{k} \cdot \vec{r}_j} \\
&= h_i \sum_{\vec{k}} \alpha_{\vec{k}} \langle \tilde{S}_{\vec{k}}^x \tilde{S}_{-\vec{k}}^x \rangle e^{i\vec{k} \cdot (\vec{r}_j - \vec{r}_i)} \\
&= h_i G_{\perp}(\vec{r}_j - \vec{r}_i)
\end{aligned} \tag{2}$$

Where we use the definition, $\alpha_{\vec{k}} = (-1)^0 + (-1)^1 e^{-i\vec{k} \cdot \vec{\delta}}$, with $\vec{\delta}$ being the bond directional vector. The choice of any of the six directional vectors around a particular spin should not affect the overall result. In the following we choose the direction $\vec{\delta} = (1, 0)$ for our calculation. Following Wollny, the momentum space response function at $T = 0$ energy can now simply be written as,

$$\begin{aligned}
G_{\perp}(\vec{k}) &= \alpha_{\vec{k}} \langle \tilde{S}_{\vec{k}}^x \tilde{S}_{-\vec{k}}^x \rangle = S \frac{\alpha_{\vec{k}}}{\gamma_{\vec{k}} - 1} \\
&\text{with,} \\
\alpha_{\vec{k}} &= 1 - e^{-ik_x} \\
\gamma_{\vec{k}} &= \frac{1}{3} \left(\cos k_x + 2 \cos(k_x/2) \cos(\sqrt{3}k_y/2) \right)
\end{aligned} \tag{3}$$

The principle contributions for the response function in the zero energy limit will come from the goldstone moders around the ordering wave vectors for triangular lattice, $\vec{Q} = (0, 0)$ and $\pm (4\pi/3, 0)$. We expand around these points, $\vec{k} = \vec{q} + \vec{Q}$ with a polar parametrization $\vec{q} = (q \cos \phi, q \sin \phi)$ and find that to leading order,

$$G_{\perp}(\vec{q}) = -\frac{4i \cos \phi}{q} + \mathcal{O}(q^0) \tag{4}$$

In the continuum language the response function in real space therefore is,

$$\begin{aligned}
G_{\perp}(\vec{r}) &= \frac{1}{2\pi} \int_0^{\Lambda} q dq \int_0^{2\pi} d\phi \left(-\frac{4i \cos \phi}{q} \right) e^{iqr \cos(\theta-\phi)} \\
&= \frac{-4i}{2\pi} \int_0^{2\pi} d\phi \cos \phi \left[\frac{e^{iqr \cos(\theta-\phi)}}{ir \cos(\theta-\phi)} \right]_0^{\Lambda} \\
&= \frac{-2}{\pi r} \int_0^{2\pi} d\phi \cos \phi \left(\frac{e^{i\Lambda \cos(\theta-\phi)} - 1}{\cos(\theta-\phi)} \right) \\
&= \frac{-2}{\pi r} \int_0^{2\pi} d\zeta \cos(\theta-\zeta) \left(\frac{e^{i\Lambda \cos \zeta} - 1}{\cos \zeta} \right) \\
&= \frac{-2 \cos \theta}{\pi r} \int_0^{2\pi} d\zeta (e^{i\Lambda r \cos \zeta} - 1) + \cancel{\frac{-2 \sin \theta}{\pi r} \int_0^{2\pi} d\zeta \sin \zeta \left(\frac{e^{i\Lambda \cos \zeta} - 1}{\cos \zeta} \right)} \\
&= \frac{4 \cos \theta}{r} [1 - J_0(\Lambda r)]
\end{aligned} \tag{5}$$

Where $J_0(x)$ is the bessel function of first kind with order zero.

3 Randomly distributed impurities and breakdown of the 120° order

The transverse fluctuation of the local magnetic moment $m_{\perp}(i) = \langle S_i^x \rangle$ due to the presence of a single bond impurity has been shown to be finite. It is straightforward to extend this picture for an arbitrary distribution of these bond impurities as long as the correlation between the impurity strengths is known. In the continuum description, $m_{\perp}(\vec{r}) = \int d^2 r' h(\vec{r}') G_{\perp}(\vec{r} - \vec{r}')$. Now, we calculate the sample averaged $\langle\langle \dots \rangle\rangle$, squared transverse fluctuation,

$$\begin{aligned}
\langle\langle m_{\perp}^2(0) \rangle\rangle &= \int d^2 r \langle\langle h(\vec{r})^2 \rangle\rangle [G_{\perp}(\vec{r})]^2 \\
&= h_0^2 \int_1^L r dr \int_0^{2\pi} d\theta \left(\frac{16 \cos^2 \theta [1 - J_0(\Lambda r)]^2}{r^2} \right) \\
&\simeq 16 h_0^2 \pi \log L
\end{aligned} \tag{6}$$

Which diverges in the thermodynamic limit for an uniform impurity distribution with a finite mean square moment, h_0^2 . This implies that in a thermodynamically large sample of triangular lattice with heisenberg interaction, any amount of uniformly distributed bond impurities would destroy the local magnetic order parameter by introducing divergent transverse fluctuation.

In light of the Imry-Ma argument, the correlation function of the transverse magnetization is also an interesting quantity to look as it can signal the break-

down of the long range magnetic order.

$$\begin{aligned} C_{\perp}(\vec{r}) &= \langle \langle m_{\perp}(0) m_{\perp}(\vec{r}) \rangle \rangle \\ &= \int d^2x \int d^2y \langle \langle h(\vec{x}) h(\vec{y}) \rangle \rangle G_{\perp}(\vec{x}) G_{\perp}(\vec{r} - \vec{y}) \end{aligned} \quad (7)$$

If the bond impurities are distributed randomly then we shall have, $\langle \langle h(\vec{x}) h(\vec{y}) \rangle \rangle = h_0^2 \delta^2(\vec{x} - \vec{y})$. Hence,

$$\begin{aligned} C_{\perp}(r) - C_{\perp}(0) &= h_0^2 \int d^2x G_{\perp}(\vec{x}) G_{\perp}(r\hat{e}_z - \vec{x}) - h_0^2 \int d^2x G_{\perp}(\vec{x})^2 \\ &= h_0^2 \int_1^L x dx \int_0^{2\pi} d\chi \left(\frac{4 \cos \chi [1 - J_0(\Lambda x)]}{x} \frac{4(x \cos \chi - r)[1 - J_0(\Lambda |\vec{r} - \vec{x}|)]}{r^2 + x^2 - 2rx \cos \chi} \right. \\ &\quad \left. - \frac{16 \cos^2 \chi}{x^2} [1 - J_0(\Lambda x)^2] \right) \\ &\simeq 16h_0^2 \int_1^L x dx \int_0^{2\pi} d\chi \left(\frac{\cos \chi (x \cos \chi - r)}{x(r^2 + x^2 - 2rx \cos \chi)} - \frac{\cos^2 \chi}{x^2} \right) \\ &\simeq 16h_0^2 \left(-\pi \log L - \frac{\pi L^2}{2r^2} \right) \\ &= -\frac{8\pi h_0^2 L^2}{r^2} - C_{\perp}(0) \end{aligned} \quad (8)$$

So, the correlation function has a power law decay, $C_{\perp}(r) \sim 1/r^2$.